Python Programming Practical R1UC427B

Week 1

Practical 1: Python Data Types & Variables

1. Print "Hello, World"
2. Program to declare and print different types of variables
3. Program to take user input and display its data type
4. Program to swap two variables
5. Program to assign multiple variables in a single line.

Practical 2: Operators in Python

6. Program using arithmetic operators
7. Program using relational operators
8. Program using logical operators
9. Program using assignment operators

Week 2

Practical 3: Conditional Statements

10. Program using simple if statement
11. Program using if–else statement
12. Program using nested if–else
13. Program using elif ladder

Practical 4: Looping Statements

14. Program using for loop
15. Program using while loop
16. Program using nested for loop
17. Program using break statement
18. Program using continue statement

Week 3

Practical 5: Pattern Printing

19. Program to print star pattern
20. Program to print number pattern
21. Program to print pyramid pattern
22. Print alphabet triangle pattern

23. Print Pascal's triangle pattern

Practical 6: Strings

24. Program to create and index a string
25. Program demonstrating string slicing
26. Program using built-in string methods
27. Program to check palindrome string
28. Program to find length of a string
29. Program to reverse a string
30. Program to count vowels in a string.

Week 4

Practical 7: String

31. Program to Count Occurrence of a Character
32. Program to Find Position of a Character
33. Program to Check Whether String Starts with a Word
34. Program to Check Whether String Ends with a Word
35. Program to Split a String
36. Program to Join Strings

Practical 8: Lists

37. Program to create and access list elements
38. Program demonstrating list slicing
39. Program using built-in list methods
40. Program to find sum of list elements
41. Program to find largest and smallest element in a list
42. Program to remove duplicate elements from a list
43. Program to sort a list

Week 5

Practical 9: Tuples

44. Program to create and access tuple
45. Program demonstrating tuple slicing
46. Program to convert tuple to list
47. Program to find length of a tuple
48. Program to concatenate two tuples

Practical 10: Sets and Dictionary

49. Program to perform set operations
50. Program using set methods
51. Program to create and print a dictionary
52. Program to add new key-value pairs to a dictionary
53. Program to update and modify dictionary values
54. Program to delete elements from a dictionary

Week 6

Practical 11: Functions

55. Program to define and call a function
56. Program to check prime number using function
57. Program to find greatest of three numbers using function
58. Program to find sum of digits using function
59. Program to check palindrome using function
60. Program to perform calculator operations using functions

Practical 12: Lambda Function, Importing Modules, Math Module

61. Program to create and use a simple lambda function
62. Program to find square of a number using lambda function
63. Program to add two numbers using lambda function
64. Program to import a module and use its functions
65. Program to use math module constants (pi, e)
66. Program to perform square root and power operations using math module
67. Program to find factorial using math module
68. Program to use trigonometric functions (sin, cos, tan)

Week 7

Practical 13: Creating Class and Object

69. Program to create a simple class and object
70. Program to create a student class and display details
71. Program to create an employee class and calculate salary
72. Program to create multiple objects for a class
73. Program to perform calculator operations using a class

Practical 14: Constructors

74. Program using default constructor
75. Program using parameterized constructor
76. Program to display student details using constructor
77. Program to create a bank account class with constructor to initialize balance
78. Program to create a simple destructor using __del__()

Week 8

Practical 15: Inheritance

64. Program demonstrating single inheritance
65. Program demonstrating multilevel inheritance
66. Program demonstrating multiple inheritance.
67. Program to demonstrate Hierarchical Inheritance

Practical 16: Encapsulation

68. Program demonstrating encapsulation.
69. Program to protect employee salary using encapsulation.
70. Program to demonstrate getter method for private variable
71. Program to protect employee salary using encapsulation

Week 9

Practical 17: Data Abstraction

72. Program demonstrating abstraction using abstract class
73. Program to implement abstract class with rectangle and circle.
74. Program to demonstrate student result using abstract method

Practical 18: Polymorphism

75. Program to create animal classes showing polymorphism.
76. Program to implement calculator operations using polymorphism.
77. Program to demonstrate polymorphism with different objects

Week 10

Practical 19:

78. Student Class Management System: Use class, object, constructor.
79. Library Management System : Class, Object, Methods.
80. ATM Machine Simulation: Encapsulation and Methods

Practical 20:

81. Library Management System: Use inheritance for books and members.
82. Vehicle Information System: Inheritance (Vehicle → Car/Bike)

Practical 21: Basics of NumPy

1. Program to create a NumPy array
2. Program to create a 1D and 2D NumPy array
3. Program to perform basic arithmetic operations on NumPy arrays
4. Program to find mean, median, and standard deviation of an array
5. Program to reshape a NumPy array
6. Program to find the maximum and minimum values in an array
7. Program to perform array slicing in NumPy
8. Program to create arrays using zeros(), ones(), and arange()
9. Program to perform matrix addition and multiplication
10. Program to find the transpose of a matrix

Practical 22: Basics of Pandas

1. Program to create a Pandas Series
2. Program to create a Pandas DataFrame
3. Program to read data from a CSV file using Pandas
4. Program to display first and last rows of a DataFrame
5. Program to select specific columns from a DataFrame
6. Program to filter rows based on conditions
7. Program to find summary statistics of a DataFrame
8. Program to add a new column to a DataFrame
9. Program to handle missing values in a DataFrame
10. Program to sort data in a DataFrame

Practical 23: Basics of Matplotlib

1. Program to plot a simple line graph
2. Program to create a bar chart
3. Program to create a pie chart
4. Program to create a histogram
5. Program to plot multiple lines on a graph
6. Program to add title and labels to a graph
7. Program to create a scatter plot
8. Program to customize graph color and style
9. Program to display grid in a graph
10. Program to plot data from a Pandas DataFrame

Practical 24:

1. Attendance Visualization: Plot attendance using Matplotlib.
2. Student Marks Analysis : Use NumPy to calculate average, maximum, and minimum marks of students.
3. Patient Health Records : Use Pandas to store patient data (name, age, blood pressure).